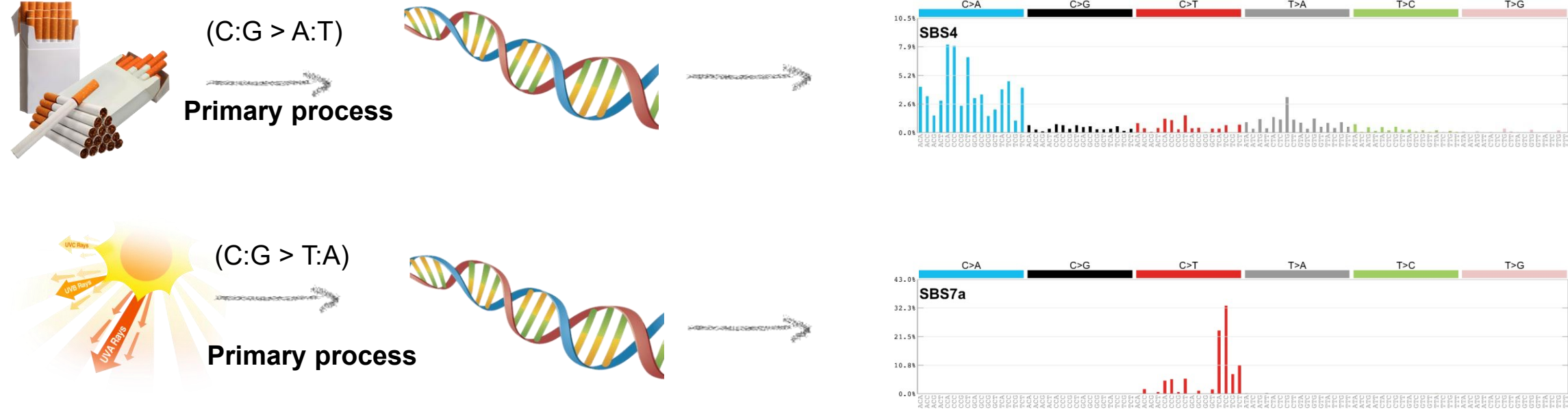


Mutational Signature Ontology & Knowledge Graph

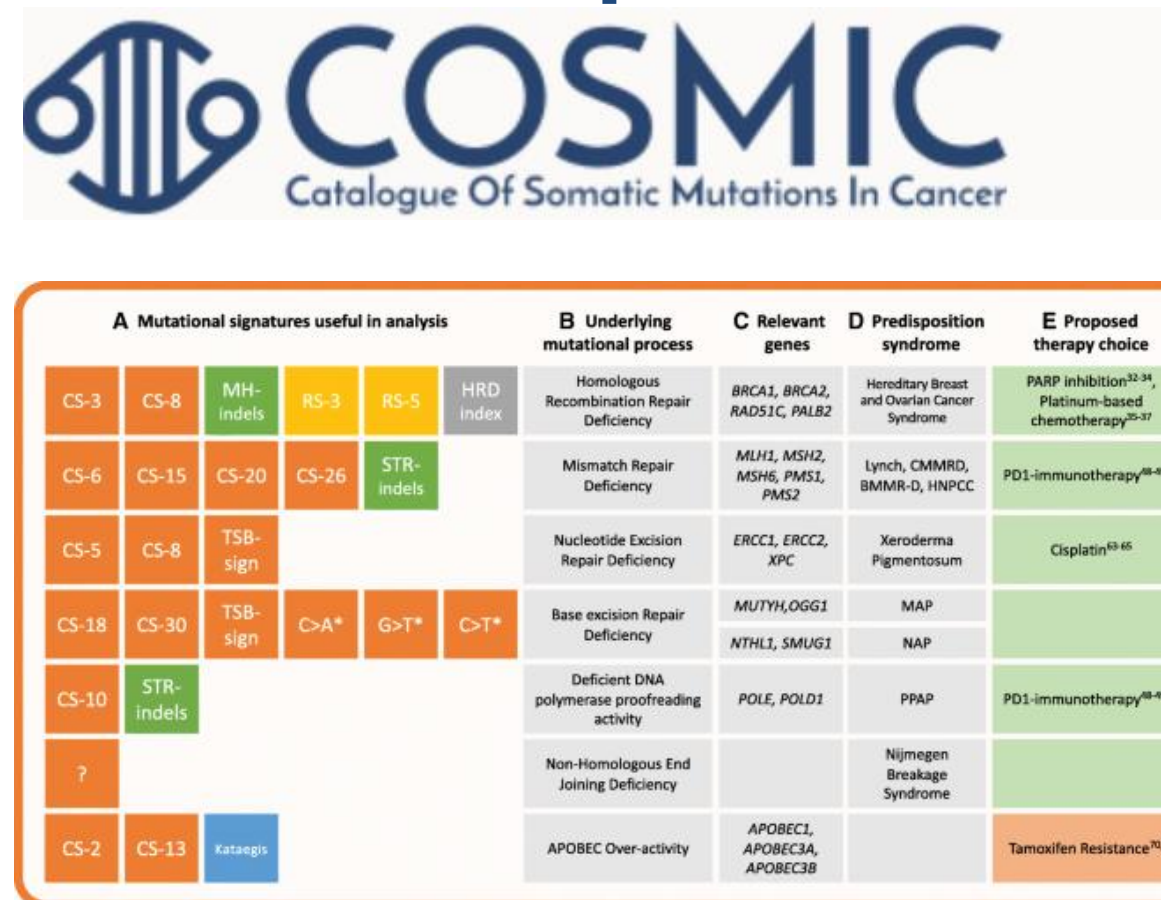
Ulrike Steindl, Kjong-Van Lehmann

Mutation processes leave distinct mutational signatures in the genome



Mutational Signatures summarize mutational patterns

- Generate meaningful subcategories in cancer diagnostics
- Determining personalized diagnoses, treatment options
- COSMIC releases a versioned catalogue of Mutational Signatures based on reference data sets



- Mutational signatures may correspond to underlying aetiologies of mutational profiles
 - Smoking
 - Homologous repair deficiency
 -

Motivation: represent complex biological relationships

- Organize and structure existing knowledge
- Facilitate automated knowledge generation in the Mutational Signatures space
- Accurately and concisely represent complex relationships
 - Mutational signatures & Cancer types
 - Mutational signatures & Treatments
 - Mutational signatures & Aetiologies
- Help data integration across multiple data sources

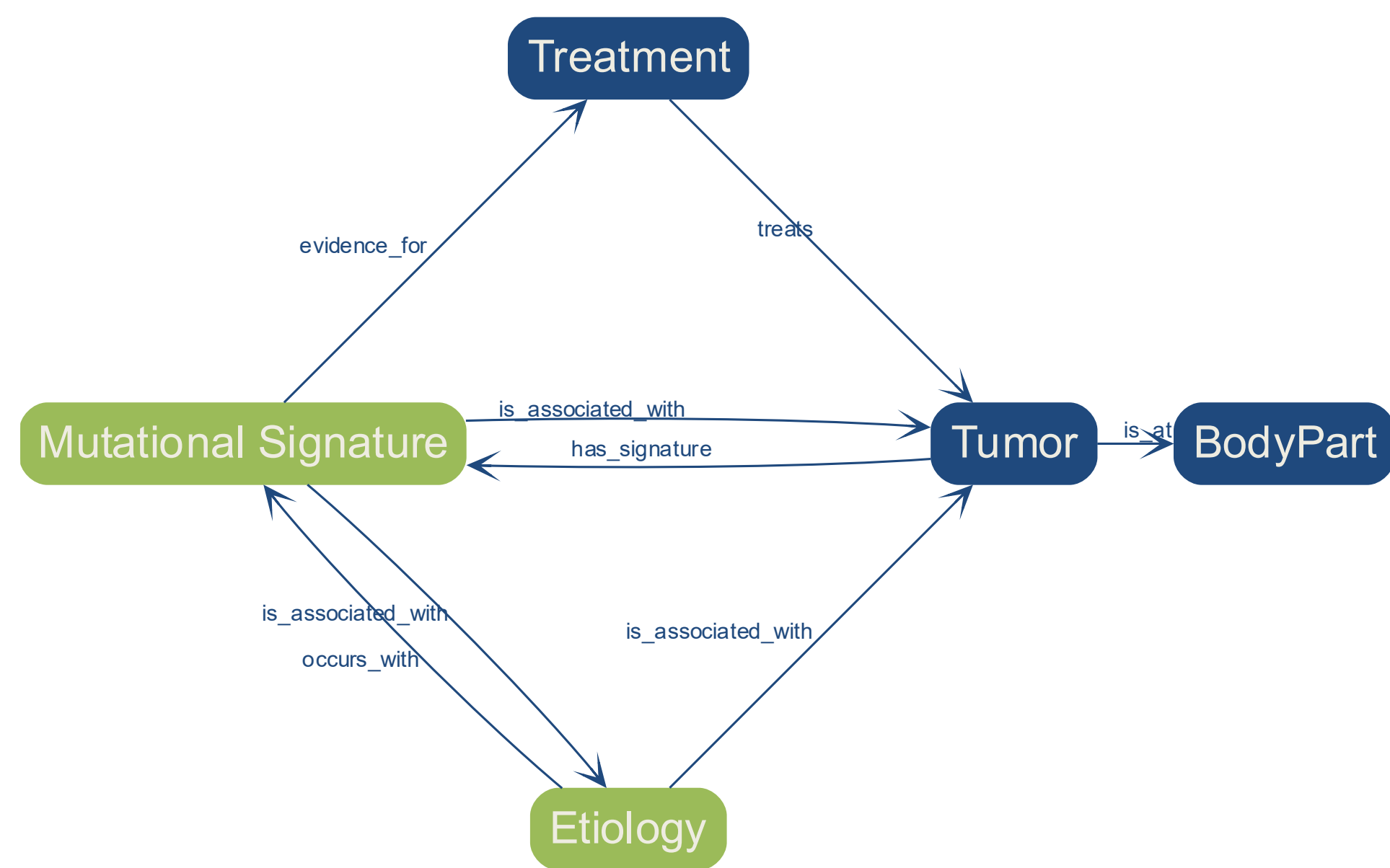
"Glioblastoma typically exhibits signatures 1, 5, 11, 12, 30, 40" (Alexandrov et al. 2020)

"Signature 3 is associated with a Homologous Recombination Repair Deficiency"

"Patients with a Homologous Recombination Repair Deficiency respond better to Platinum-based therapy" (Van Hoeck et al 2019)

Ontology design

Focus on interoperability



Owl format

Concepts

- Mutational Signature
- Treatment
- Aetiology
- Tumor
- BodyPart

Link to other Ontologies like NCIT (National Cancer Institute Thesaurus)

is_associated_with

Taxonomy for aetiologies:

- Substance Exposure
- Chemotherapy
- Mutation
- Deficiency
- Sampling artifact
- Unknown

Each Observation has

- Version
- Convention

Mutational Signatures use Non-negative Matrix Factorization

- Extracted from mutational counts for each of the 96 combinations (A[C>A]A, T[G>C]G, etc.)
- 86 signatures extracted

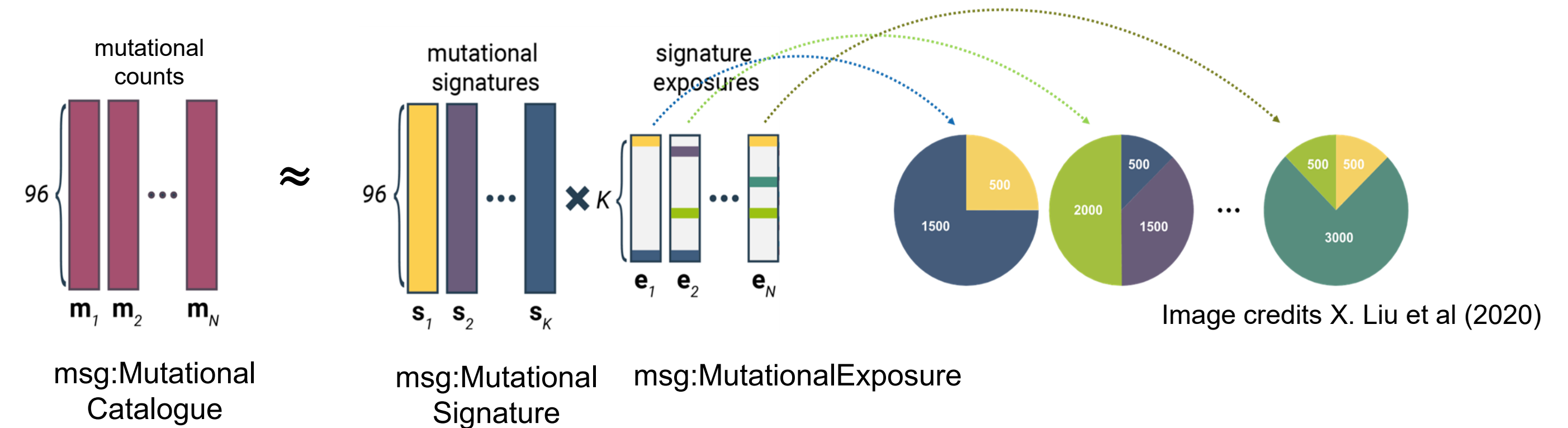
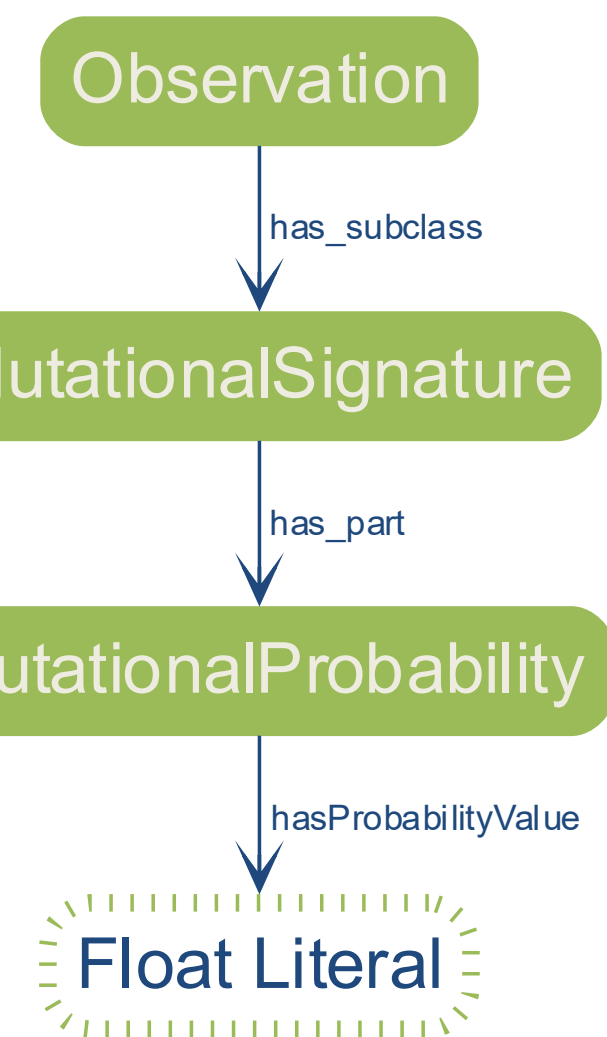


Image credits X. Liu et al (2020)

Closely model the mathematical base

Relationship between Signatures and Values

- has_part
- hasProbabilityValue



Knowledge Graph

With structured and unstructured research findings

Rdf format

Numeric values for COSMIC SBS mutational signatures

COSMIC metadata (Sondka et al. 2023)

Academic findings (Alexandrov et al. 2020)

Selected unstructured knowledge (Van Hoeck 2019)



Querying the knowledge graph

For a given mutational signature, which cancer types is it associated with?

- sbs11, Panc-Endocrine
- sbs11, CNS-GBM
- sbs12, Biliary-AdenoCA
- sbs12, Liver-HCC

Which signatures were added in version v3.1?

- sbs 86 – sbs 90

Which signatures does Bladder cancer have?

- sbs1, sbs2, sbs5, sbs8, sbs13, sbs29

Which mutational signatures have been confirmed experimentally?

- [msg:SBS_v3.4_sbs10a](https://doi.org/10.1172/jci122095), doi:10.1172/jci122095
- [msg:SBS_v3.4_sbs10b](https://doi.org/10.1172/jci122095), doi:10.1172/jci122095
- [msg:SBS_v3.4_sbs11](https://doi.org/10.1016/j.cell.2019.03.001), doi:10.1016/j.cell.2019.03.001

Current status: work in progress

- Ontology with 1,021 triplets
- Knowledge graph with 169,692 triplets
- Queryable and linked to other ontologies
- BFO / OBI / SIO: For methodological vocabulary
- Test
- Validate with SHACL Shapes
- Document
- Release

Improve data coverage

Include different mutation types

Double base substitutions

Copy number variations

Include other signature types

e.g. Degasperri et al 2020

Focus on genomic location

Aligning with BFO/SIO ontology architecture

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